

Analysis of the relationship between city size and municipal spending efficiency in California cities.*

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California's rising government debt over the past decade necessitates careful examination of municipal spending efficiency. To analyze spending efficiency I use data on expenditure per capita for all California cities over the past 20 years. I use regression analysis to model the relationship between population size(the predictor) and per capita spending(the response variable). Then hypothesis testing is used on the slope to determine whether large cities benefit from economies of scale. The results were that there was not enough statistical evidence to claim that expenditure per capita decreased as population increased which is significant because it implies that the continued expansion of large cities which serve as the State's primary economic growth engine will not result in a reduction in spending efficiency.

1 Introduction

The motivation for this paper comes from seeing the rise in government expenditure and subsequent increase in debt in the United States. This prompted me to look into ways to make government expenditure more efficient. I later found a paper which discovered that cities in Spain experienced economies of scale effects up to a certain point (Benito, Bastida, and Guillamón 2010), which led me to investigate whether those same efficiency gains existed in California. Some useful background on California state expenditure is that over the period from 2019 to 2024 state expenditure increased from around 300 billion to 494 billion. It is currently on pace to surpass peak covid spending (499 billion) by the end of the year(2020), making it crucial to determine how efficiently this money is being spent.

*Project repository available at:<https://github.com/tycebot/Modeling-population-and-spending>.

The knowledge gap being addressed is understanding the relationship between population and expenditure on the city level in California. Most of the analysis of expenditure focuses on a subset of expenditure (like health care or education). What this paper looks to do is examine expenditure on the more granular city level with the hope that measures of spending efficiency on the city level translate to the state level. In order to address the knowledge gap the simple linear regression model was fit to measure the relationship between estimated population(the predictor) and expenditure per capita(response). Then a one sided hypothesis testing was done on the slope of the regression model to determine whether large cities experience economies of scale in their spending.

The structure of the project is divided into four sections Section 2,Section 3,Section 4, and Section 5. Section 2 introduces the data used in this analysis including limitations,definitions, and why the data is relevant. Section 3 explains the statistical methods used for the analysis. Section 4 discusses the results of the analysis as well as what can be gleaned from these results. Section 5 provides an overview on the project before going into the results. It also addresses potential shortcomings and any future extensions.Finally Section 5 contains the citations for any external sources used in the paper.

2 Data

The California capital expenditure data set is provided by the California State Controller's Office(Matthew 2025). Each row in this data set represents the expenditure per capita of a particular city in California over a fiscal year. The key variables are entity name which is the name of the city,fiscal year(July 1 of the current year to June 30 of the next year) which is the year each expenditure per capita is calculated, total expenditure which is the gross expenditure of a particular city over the fiscal year, expenditures per capita which is total expenditures divided by estimated population, and estimated population which is the population of the city based on the Housing Unit Method. The Housing Unit Method extrapolates population by taking housing units and adjusting for occupation then combines that with estimates for "group quarters population"(essentially everyone in nontraditional households like dorms and prisons) to get the overall population estimate(more details in reference). The data is limited to the fiscal year 2019 to reduce temporal dependence and because it is the most recent non-covid year with records for every city. Other limitations for the data set are that the estimated population does a bad job of measuring the homeless population(because of it's reliance on housing units) and the estimated population can double count those that move until they register their new address. By analyzing how much money is spent by a city over a fiscal year for cities with a variety of different populations we can reasonably ascertain the relationship between expenditure and the population. The key assumptions for this data set is that the population measurement is accurate relative to other cities in the data set(meaning if it underestimates or overestimates the population then it does it consistently) and that the cities are correctly reporting their expenditures to the state.

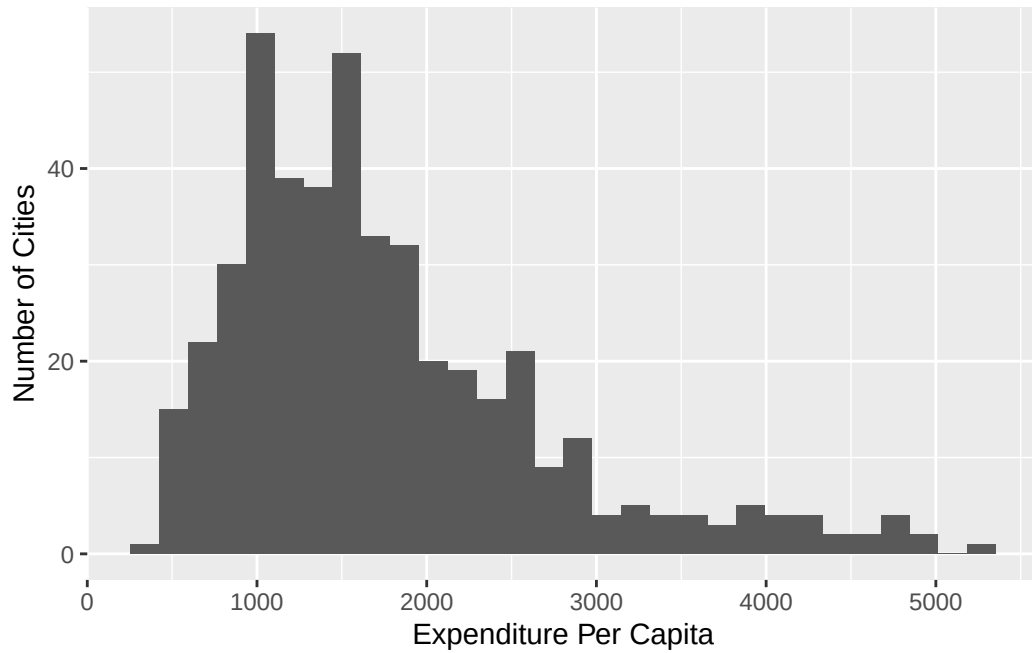


Figure 1: Histogram of average expenditure of each city which appears approximately normal but right skewed.

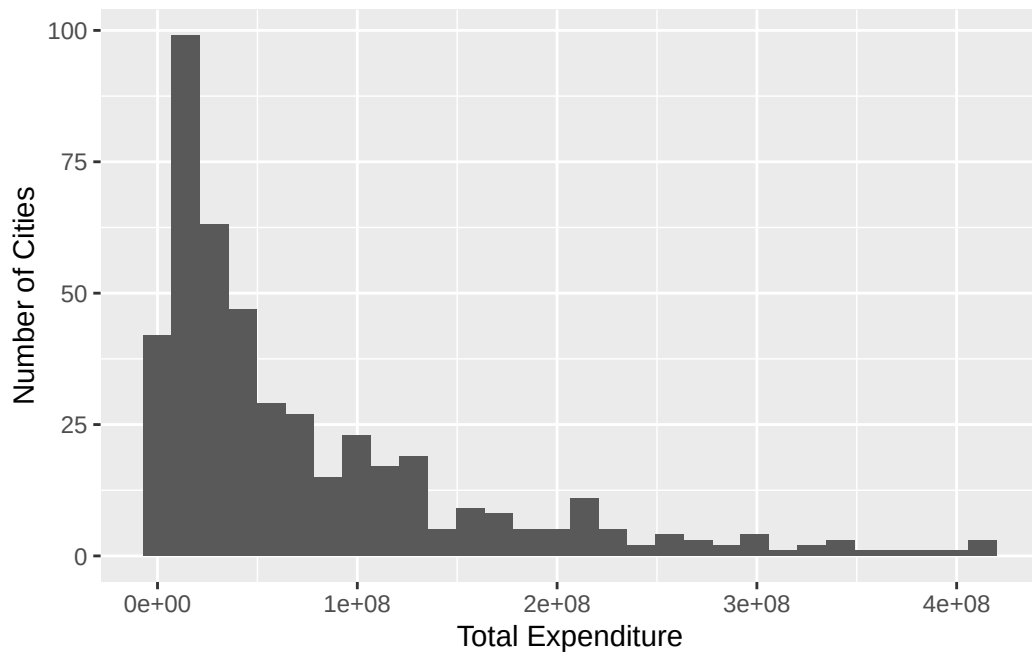


Figure 2: Histogram of total expenditure of each city which is heavily right skewed

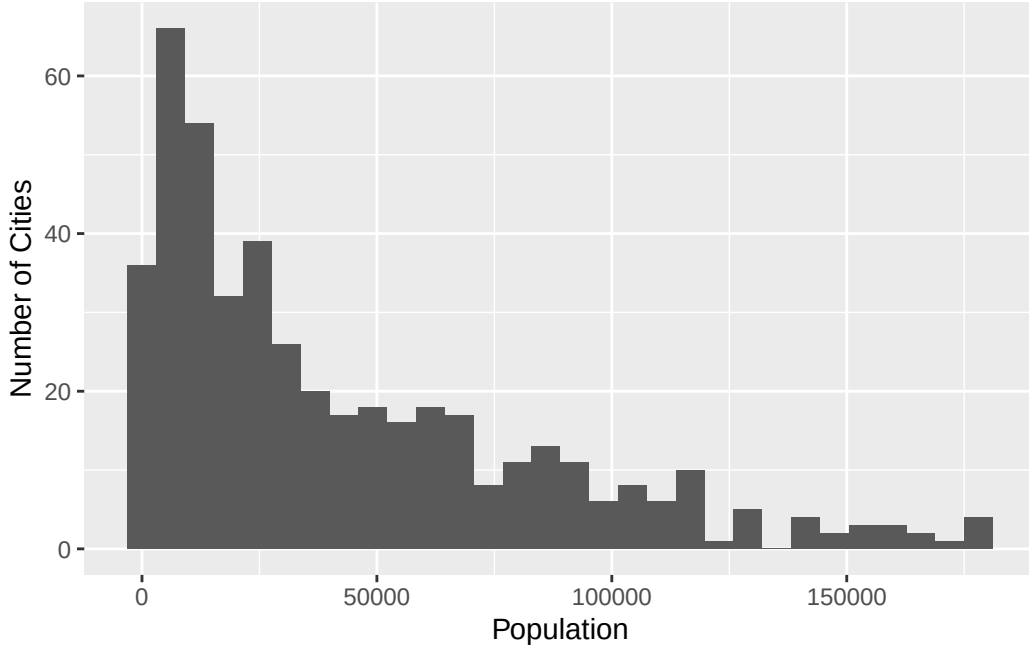


Figure 3: Histogram of population of each city which is heavily right skewed.

3 Methods

We adopt a simple linear regression model with expenditure per capita as the response and population as the predictor. Let Y_i denote expenditure per capita and X_i denote population for the i th observation in our dataset. The model can be written as $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$ where β_0 represents the expected expenditure when the population is 0, β_1 represents the average increase in expenditure per capita for every individual added to the population, and ϵ_i represents the change in expenditure per capita not captured by the population. The `lm` function is the function used to create linear models.

There are three key assumptions about the errors which allows us to use a t distribution for hypothesis testing the coefficients. The assumptions are that ϵ_i are independent of each other, have constant variance $Var(\epsilon_i) = \sigma^2$, and are normally distributed $\epsilon_i \sim N(0, \sigma^2)$. In order to evaluate whether or not high population cities get an economies of scales bonus we use a one sided hypothesis test to determine whether there is a negative association between expenditure per capita (Y) and population (X). The type 1 error rate will be set at 0.05 and the null and alternative hypothesis are as follows

$$H_0 : \beta_1 \geq 0$$

$$H_1 : \beta_1 < 0$$

I implemented this analysis using the R programming language(R Core Team 2025) and the tidyverse package(Wickham et al. 2019) for data cleaning and plot creation.

4 Results

The estimated slope parameter is $b_1 = -0.005$. In other words, for each individual added to the population the expected change in the expenditure per capita is -0.007. The estimated intercept is $b_0 = 5832.606$. In other words the expected expenditure per capita of a city with a population of zero is 5832.606.

Since the p value 0.34 is greater than the significance level($\alpha=0.05$) we fail to reject the null hypothesis that the slope is greater than or equal to 0.

The model and hypothesis test rely on 5 key assumptions which are that the regression function is linear, the error terms have equal variance, the error terms are independent, and the error terms are normally distributed. Based on the scatter plot of population against average expenditure the regression function does not seem to be linear. Equal variance does seem to hold for most of the fitted values but the range of residuals for fitted values below 1400 definitely seems smaller which violates the assumption of equal variance. Based on the curve pattern in the Q-Q plot it can be inferred that the error terms are not normally distributed. Based on the up and down pattern in the residuals vs index plot it can be inferred that the errors are not independent. Since almost all the assumptions necessary for hypothesis testing are not met we cannot trust the outcomes from the inference on the slope because we can not trust that b_1 is an unbiased estimator of β_1 .

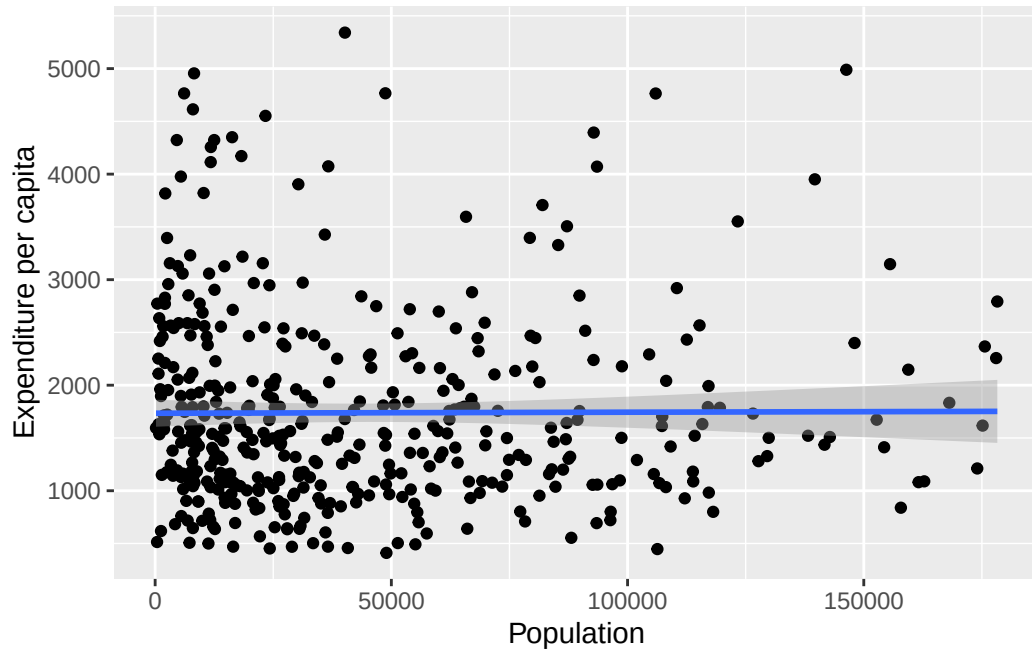


Figure 4: Scatter plot of population (x-axis) and expenditure per capita (y-axis) for a simple linear regression with expenditure per capita as the response and population as the predictor. With a straight trend line showing no relationship between population and expenditure. The confidence bands show that the variance of expenditure per capita is larger at higher populations.

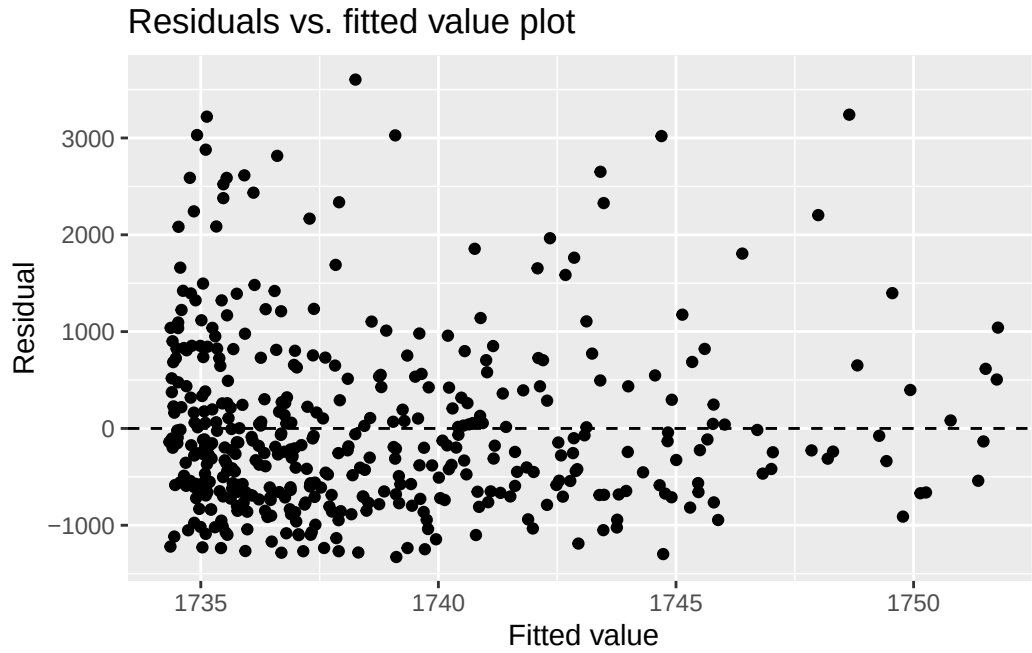


Figure 5: Scatter plot of fitted values (x-axis) and residuals (y-axis) for a simple linear regression with expenditure per capita as the response and population as the predictor.

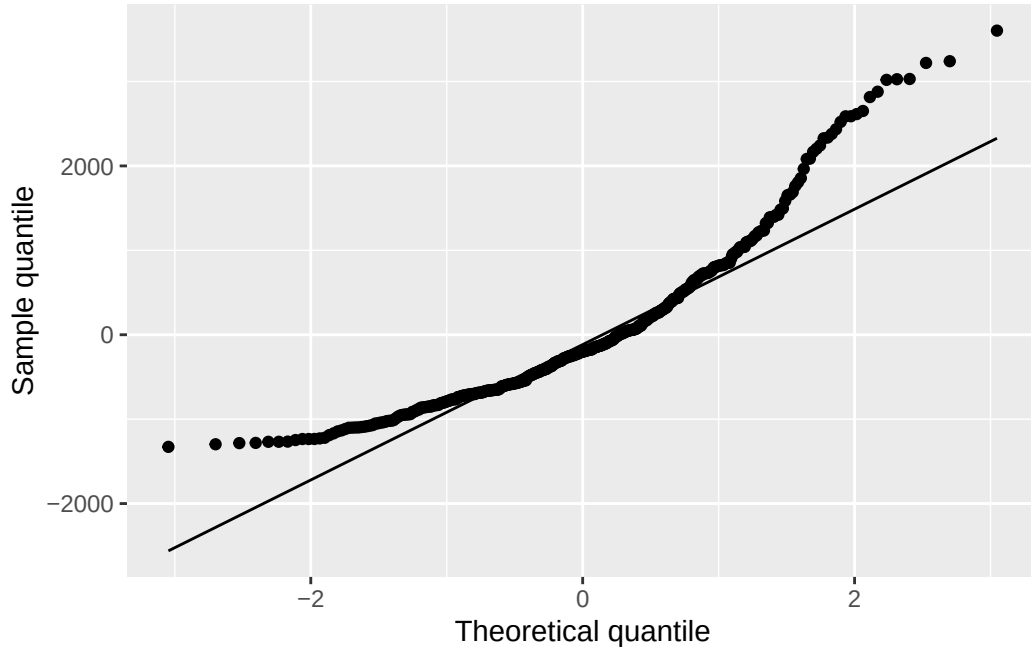


Figure 6: Quantile-quantile plot of residuals for a simple linear regression with expenditure per capita as the response and population as the predictor. With the diagonal line representing what it would look like if the sample quantile perfectly matched up with the Sample quantile. The deviation of the points from the theoretical quantile suggest the residuals aren't normally distributed.

5 Discussion

In conclusion this paper was looking to examine whether large cities benefited from economies of scale bonuses in 2019. It did this by looking at a data set of expenditure per capita and population for every city in California over a 20 year period from 2003 to 2023. A regression model was then fit to this data with expenditure per capita as the response variable and population as the predictor which yielded a slope of -0.005. A one sided hypothesis test was then conducted on the slope with a type 1 error rate of 0.05 yielding a p-value of 0.34. From the results of our regression model and hypothesis test we can conclude that there is not enough statistical evidence to reject the null hypothesis that the slope is greater than or equal to zero.

The key finding from my research was that there is not a significant economies of scale bonus granted to larger cities when spending. Although based on the slope there also does not seem to be a positive association between population and expenditure per capita. This suggest that there is not a linear association between population and expenditure per capita. The implications of this are two-fold. Firstly mandatory new costs, like added bureaucracy, that is

necessitated when administering a larger city does not seem to have a particularly negative impact on city expenditure. Second the economies of scale one would assume a larger city would gain (and which were found in a previous paper on the subject in Spain (Benito, Bastida, and Guillamón 2010)) do not exist in California. Ultimately the results suggest that the continued movement of people to large cities should not necessarily have a negative impact on spending efficiency.

There are a few key weaknesses to this study. First, because of the scale of California's major cities there are a three huge outliers when it comes to population. These outliers (Los Angeles, San Diego, San Francisco) all have populations over a million while the median population of a city in California in 2019 was 30884. Therefore they have an out sized impact on the relationship between population and per capita expenditure. Second, because the population is calculated by residents it makes it so that cities like Vernon which mainly serve as hubs for business and industry can have a relatively meager population (301) but very significant expenditure (848814). This points to a broader problem with using expenditure per capita as a proxy for spending efficiency which is that it completely ignores revenue. Thus all conclusions about expenditure's impact on spending efficiency make the assumption that revenue scales with population which is a very strong assumption.

Potential improvements that can be done to this study is combine the expenditure data set with the revenue data set. We could then infer the profit from the difference between the revenue and expenditures and use that to better measure the relationship between population and fiscal health. Potential extensions of this work would be to apply the profit analysis mentioned previously to states outside of California. The results could then be compared to see if the lack of an economies of scales bonus was due to factors specific to California or more widely common. Finally combining population with some kind of economic metric (possibly gdp per capita) would do a better job at answering the broader question of as a city grows (both economically and in population) how does that effect the health of it's finances.

Works Cited

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